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Task 5

Multi-Factor Authentication comparison Report

MFA Comparison Report

● **Introduction**

Multi-Factor Authentication (MFA) is a security method that requires a user to provide **two or more different authentication factors** before getting access to an account or system.

MFA provides stronger security than using only a password because even if an attacker knows the password, they may still be unable to access the account without the additional authentication factor.

Example:

A user logs into an account using a password and then enters a one-time password (OTP) received on their mobile phone. The password and OTP provide two steps of verification.

● **What is Authentication?**

Authentication is the process of **verifying the identity of a user** before allowing access to a system, application, or account.

For example, when a person enters their username and password to log into an email account, the system checks whether the provided credentials are correct.

● **What is Multi-Factor Authentication?**

Multi-Factor Authentication is an authentication method that uses **two or more different types of authentication factors**.

The main authentication factors are:

1. **Knowledge – Something you know**
2. **Possession – Something you have**
3. **Inherence – Something you are**

Using multiple factors makes unauthorized access more difficult.

Knowledge Factor – Something You Know

A knowledge factor is information that only the user is expected to know.

Examples:

- Password
- PIN
- Security question
- Passphrase

Advantages:

- Easy to use
- Low cost
- Simple to implement

Disadvantages:

- Can be guessed
- Can be stolen through phishing
- Users may reuse passwords

Example:

A user enters their password to log into an online banking account.

2 Possession Factor – Something You Have

A possession factor is something that the user physically possesses.

Examples:

- Mobile phone
- Security token
- Smart card
- Authentication app
- Hardware security key

Advantages:

- Provides an additional security layer
- Difficult to access without the physical device
- Useful with password-based authentication

Disadvantages:

- Device can be lost or stolen
- Some methods such as SMS OTP can be vulnerable to attacks such as SIM swapping

Example:

After entering a password, a user receives an OTP on their registered mobile phone and enters it to complete login.

3 Inherence Factor – Something You Are

An inherence factor is based on the user's unique physical or biological characteristics.

Examples:

- Fingerprint
- Face recognition
- Iris recognition
- Voice recognition

Advantages:

- Convenient
- Difficult to copy exactly
- No password needs to be remembered

Disadvantages:

- Requires compatible hardware
- Biometric data needs strong protection
- If biometric information is compromised, it cannot be changed like a password

Example:

A smartphone allows access after verifying the user's fingerprint.

- **Example of MFA in Real Life**

Consider an online banking account.

Step 1: The user enters their username and password.

Step 2: The bank asks for an additional verification method, such as an OTP or authentication-app code.

Step 3: The user provides the additional verification.

Step 4: The bank verifies both factors and allows access.

Here, the password represents the **knowledge factor**, while the mobile device/authenticator represents the **possession factor**.

● Comparison of Authentication Factors

Factor	Meaning	Examples	Main Advantage	Main Risk
Knowledge	Something you know	Password, PIN	Easy and inexpensive	Can be stolen or guessed
Possession	Something you have	Phone, OTP app, security key	Adds another security layer	Device can be lost or stolen
Inherence	Something you are	Fingerprint, face, iris	Convenient and difficult to replicate	Biometric data must be protected

Conclusion

Multi-Factor Authentication is an important security technique that provides stronger protection than password-only authentication. It uses different authentication factors such as **something you know, something you have, and something you are**.

Using MFA can significantly reduce the risk of unauthorized account access, especially when passwords are compromised. Organizations and individuals should enable MFA wherever possible and choose secure authentication methods appropriate for the level of risk.

Interview Questions

1. What is MFA?

MFA (Multi-Factor Authentication) is a security method that requires **two or more different authentication factors** to verify a user's identity.

Example: Password + OTP.

2. What are authentication factors?

There are three main authentication factors:

- **Something you know** — Password/PIN
- **Something you have** — Phone/Security key
- **Something you are** — Fingerprint/Face

3. Is OTP alone MFA?

No. OTP alone is only one authentication factor. It becomes MFA when combined with a different factor, such as **Password + OTP**.